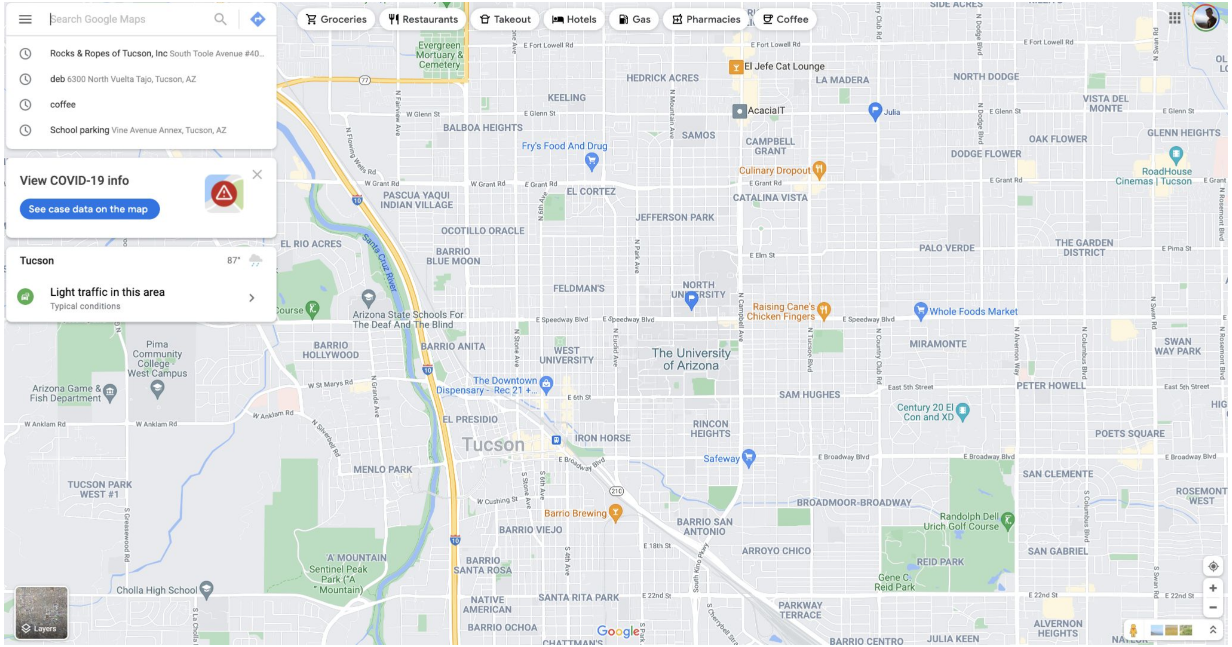
Principles of visualization

An attempt to demystify

What is a visualization?

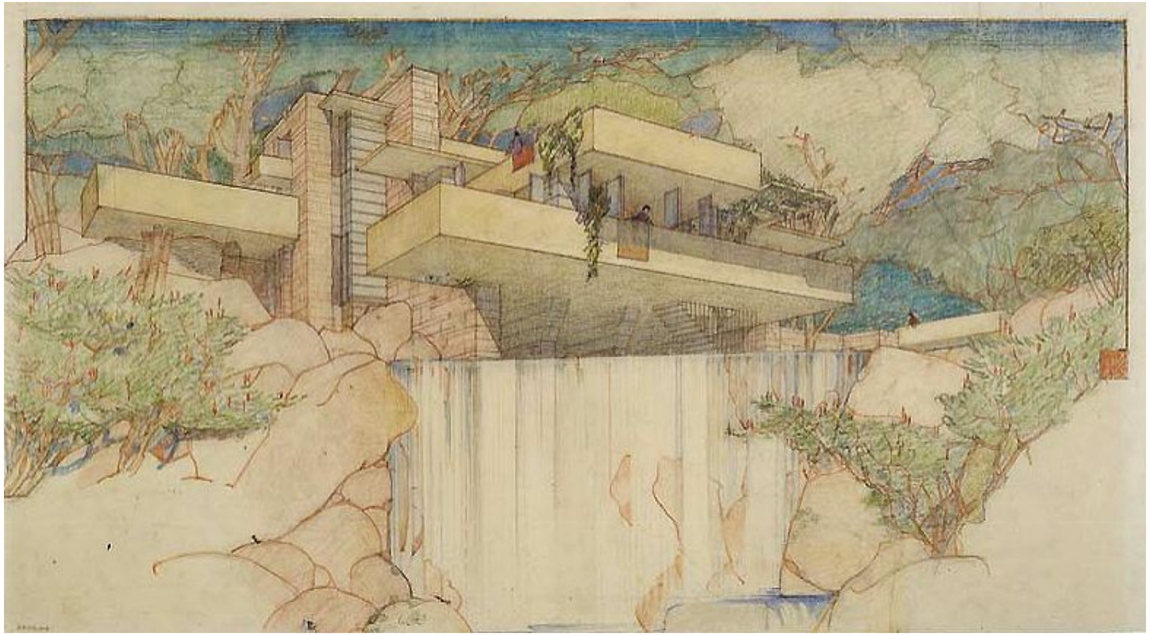
A visualization is a model of the world

The saying “All models are wrong, some are useful” applies here as well



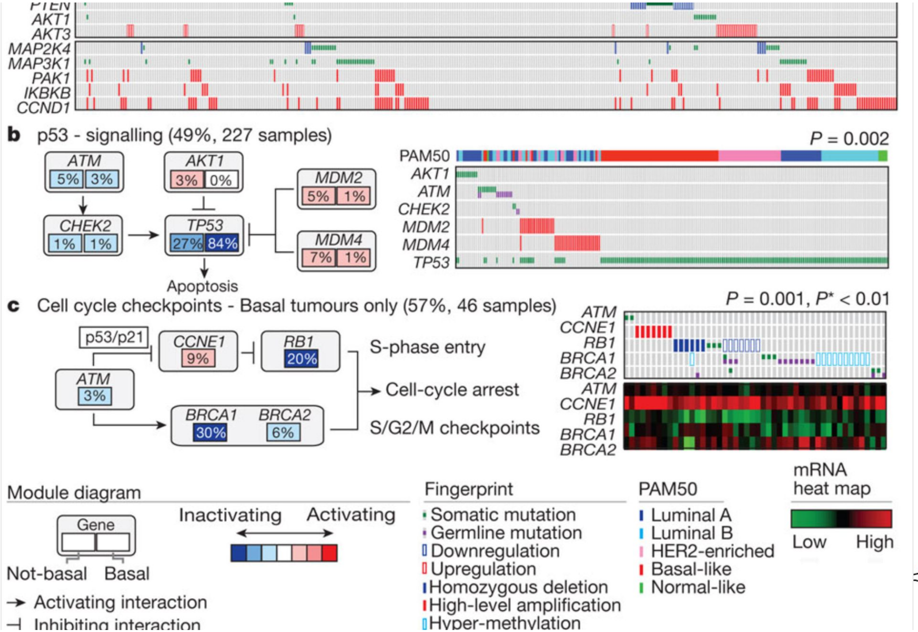
A visualization is a form of communication

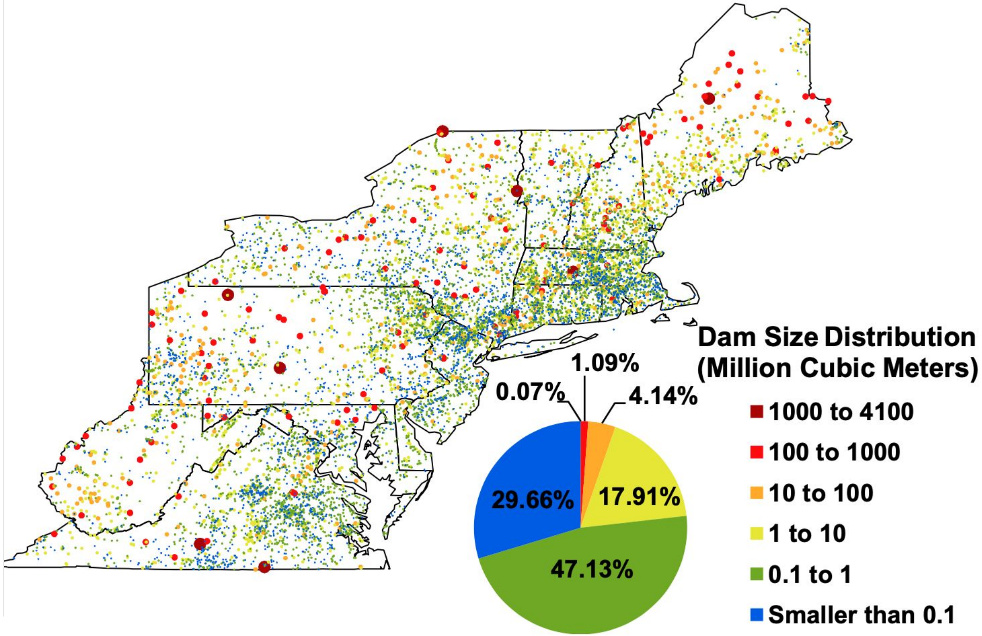
Ask yourself "What am I trying to communicate and why"



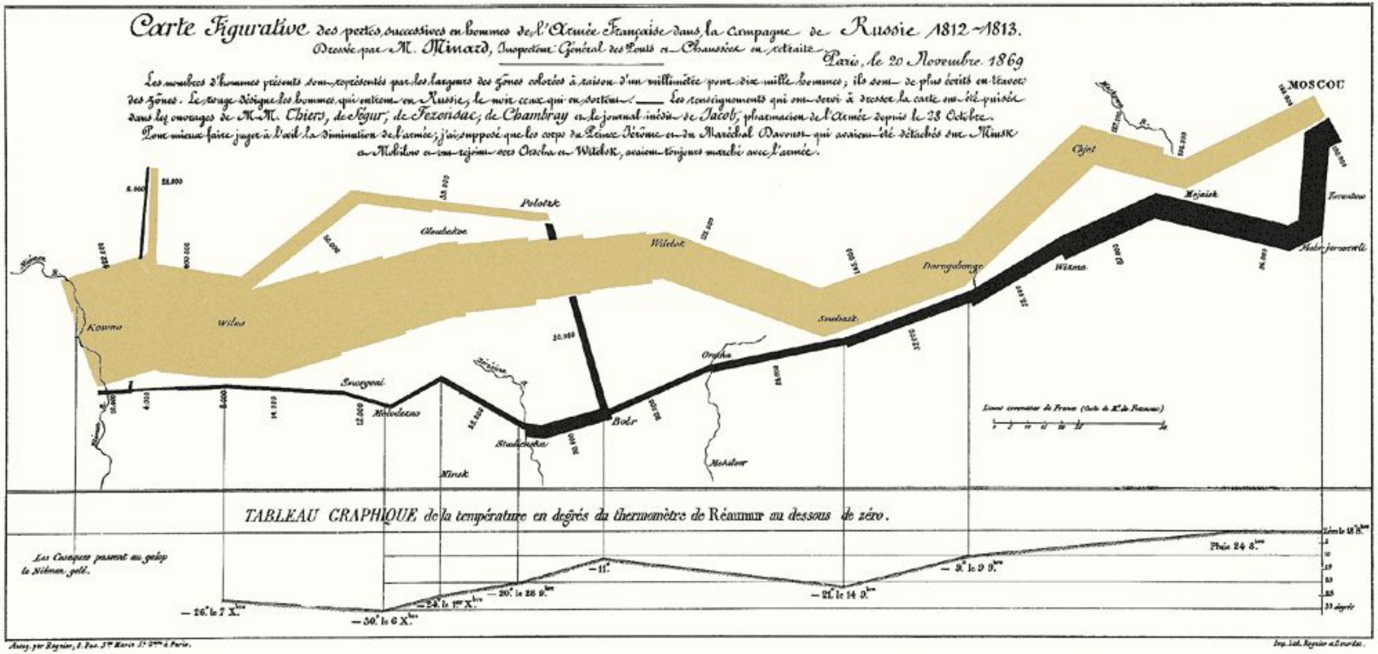
What Makes a Good Visualization?

Which do you like better? Why?





What makes a good visualization?



What are Some Components of Good Visualization?

1. Intuitive
   a. Uses visual processing well
2. Understandable
   a. Links directly to the real world
3. Accurate
   a. Not trying to mislead

Visual ingredients

The tools available to us

There are four main visual ingredients

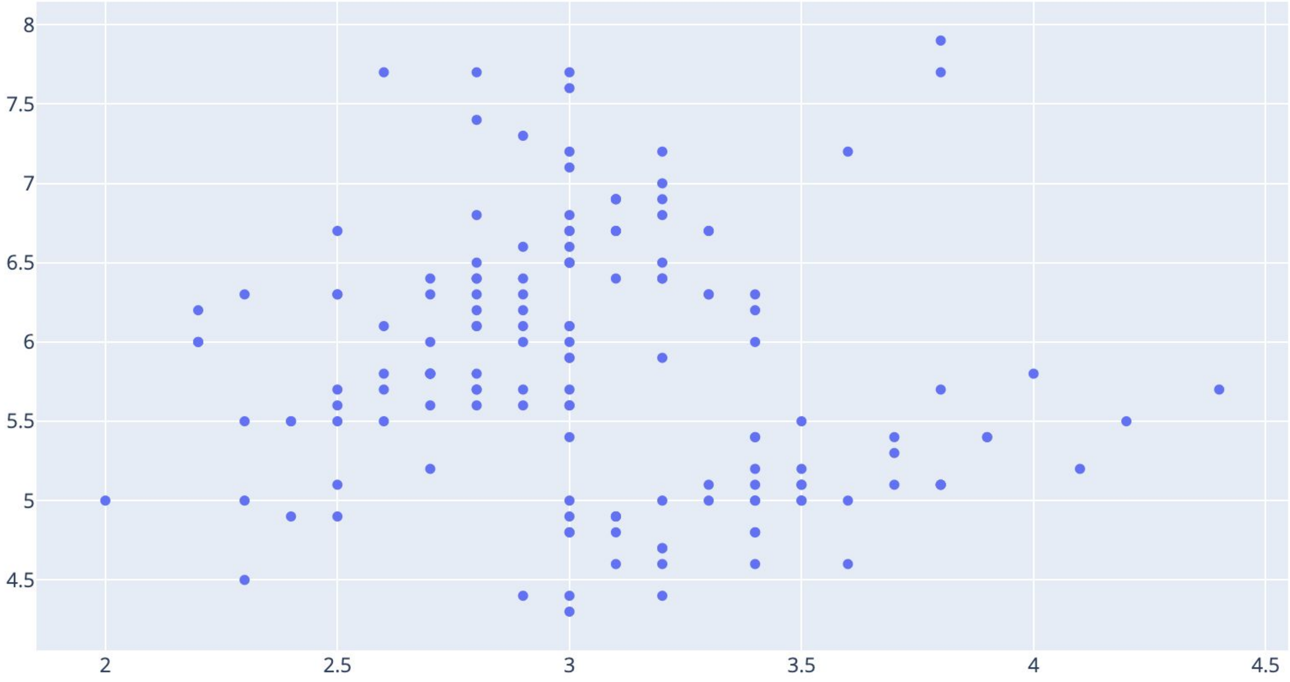
Shape

Color

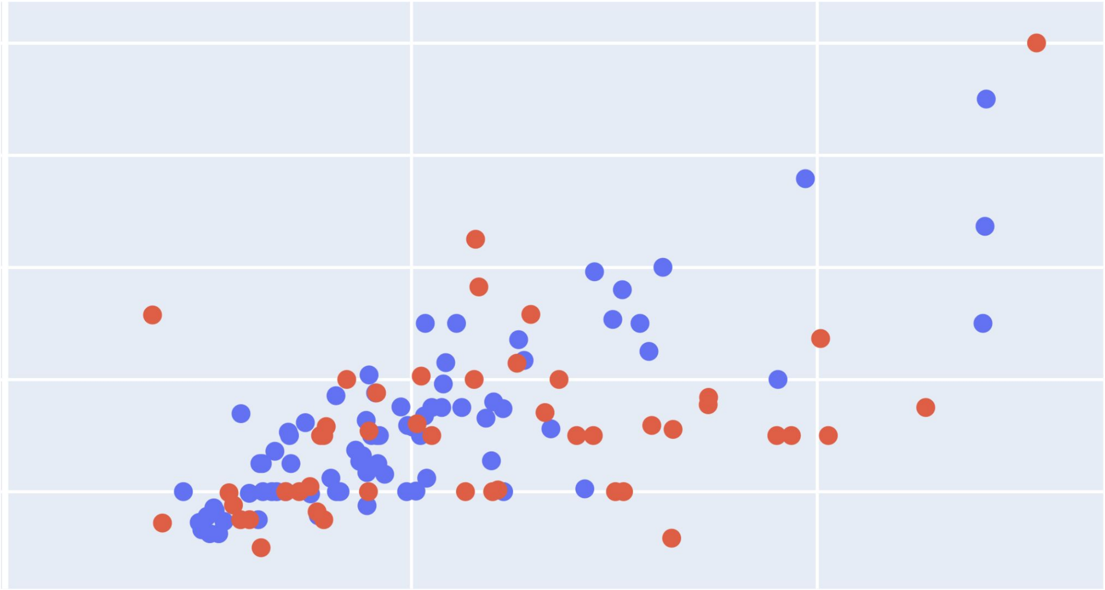
SIZE

Location

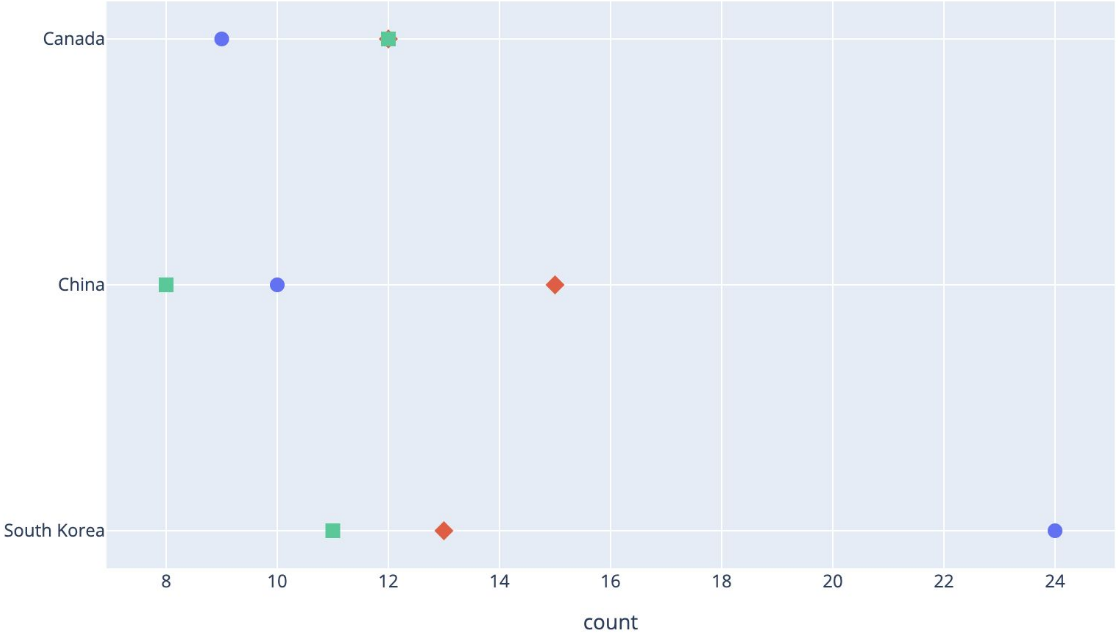
Location



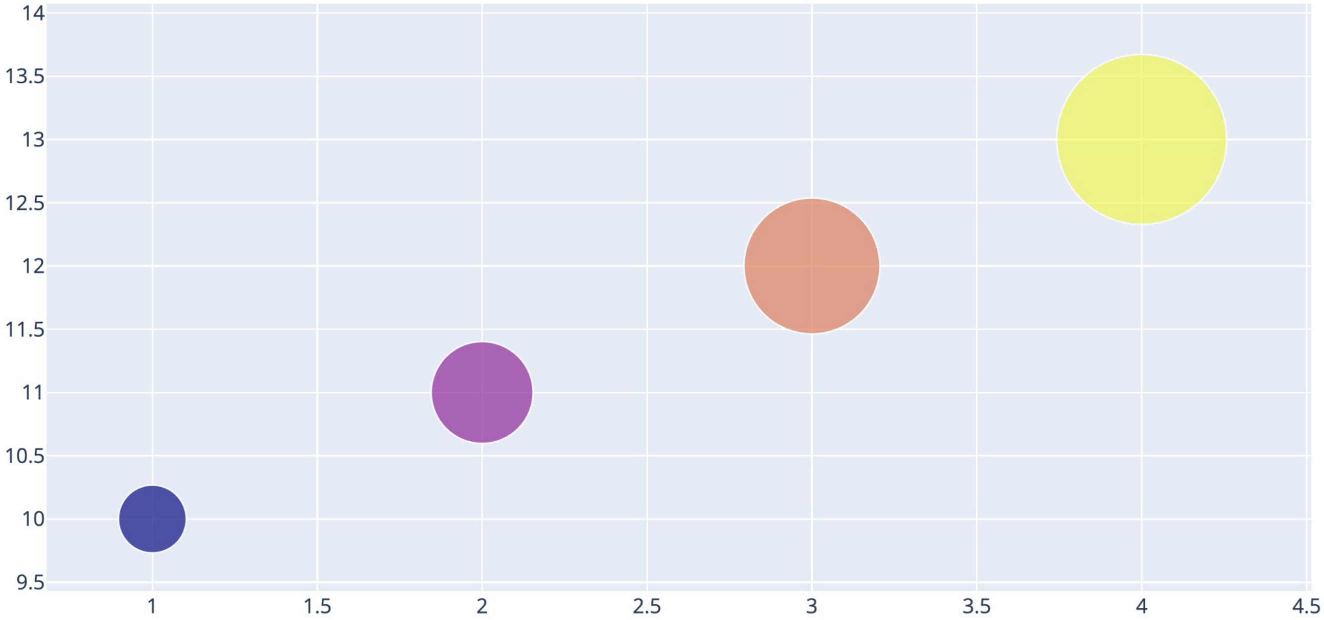
Color



Shape

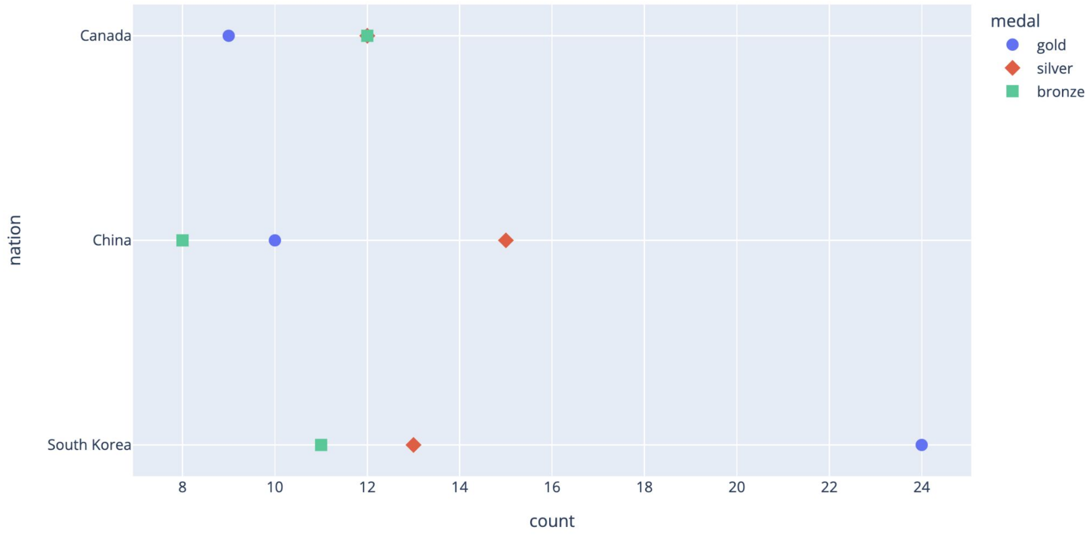


Size

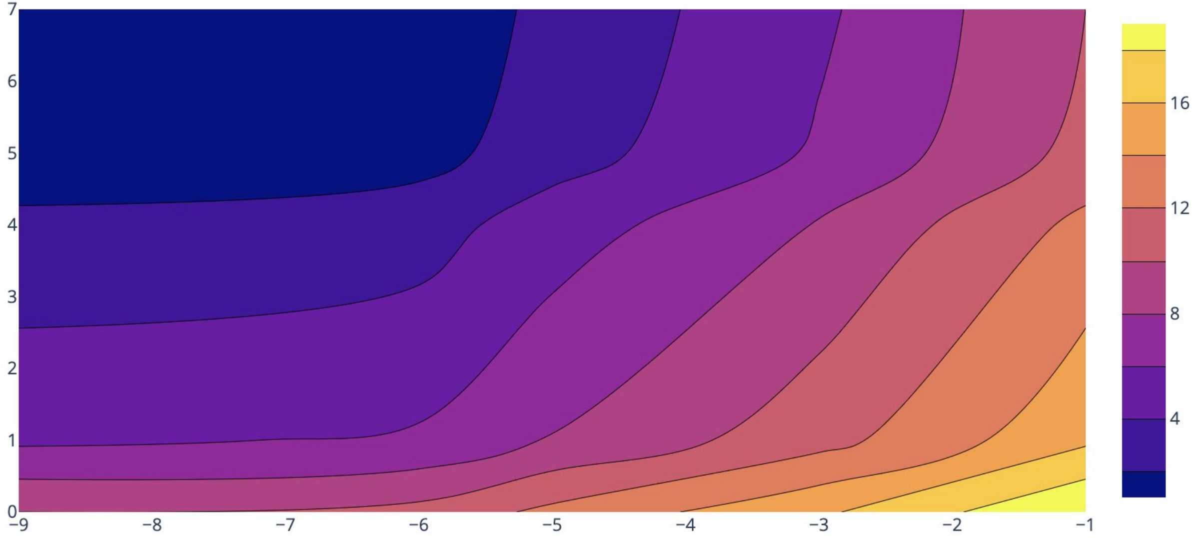


As a rule each data dimension should correspond to one visual dimension

This example is NOT doing that



This example IS doing that



Each visual ingredient has its own associations

You want the associations to align well with the qualities of the data you are trying to visualize

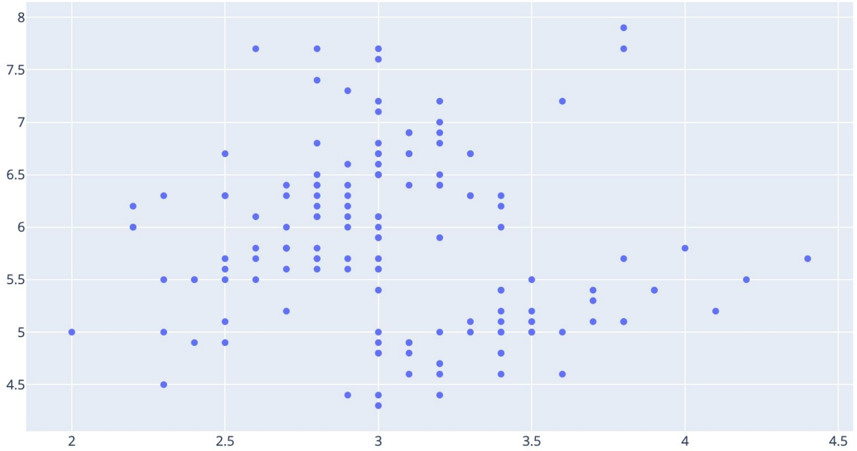
So what kinds of data are there?

Quantitative

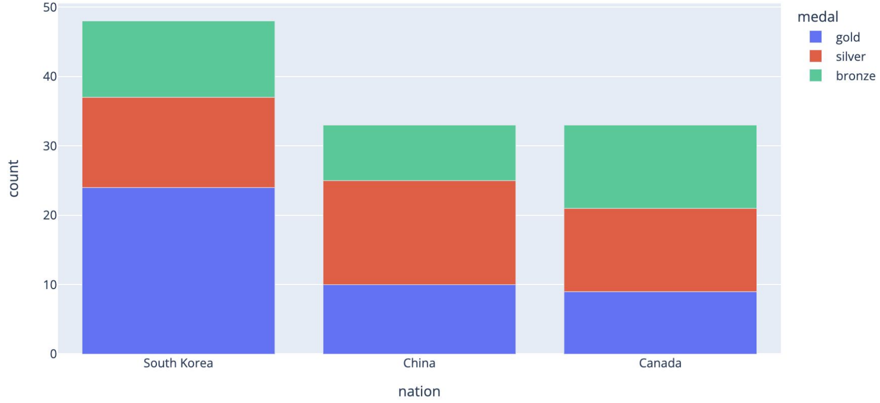
Categorical

- Nominal
- Ordinal
- Interval

Quantitative vs. categorical data



Quantitative



Categorical

Which ingredient goes to which?

Important distinction is that quantitative data is continuous

Quantitative

L o c a ti o n

Color

Size

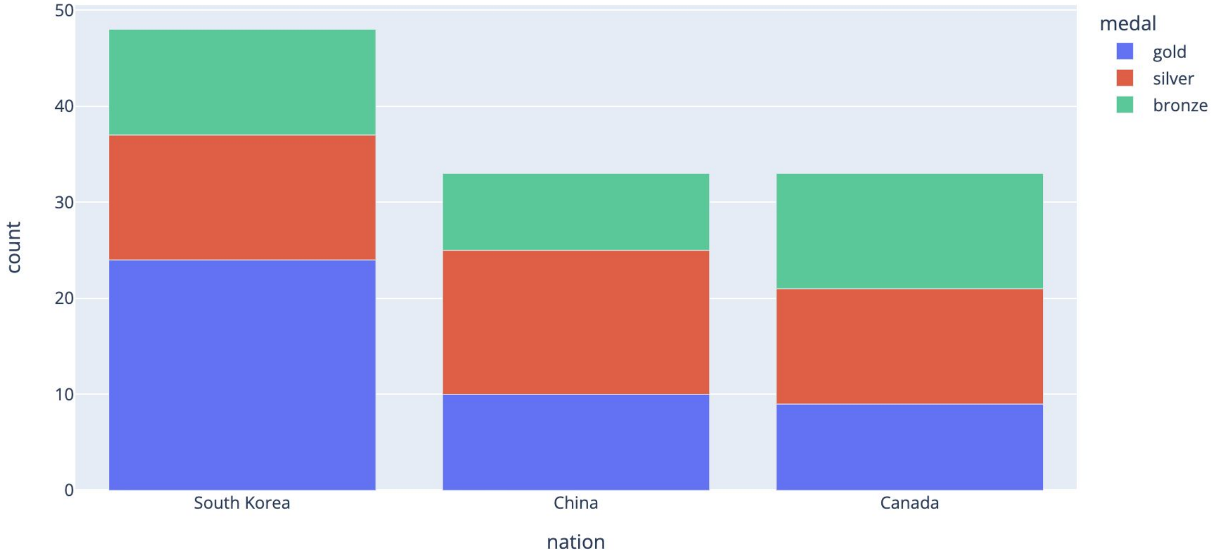
Categorical

Location

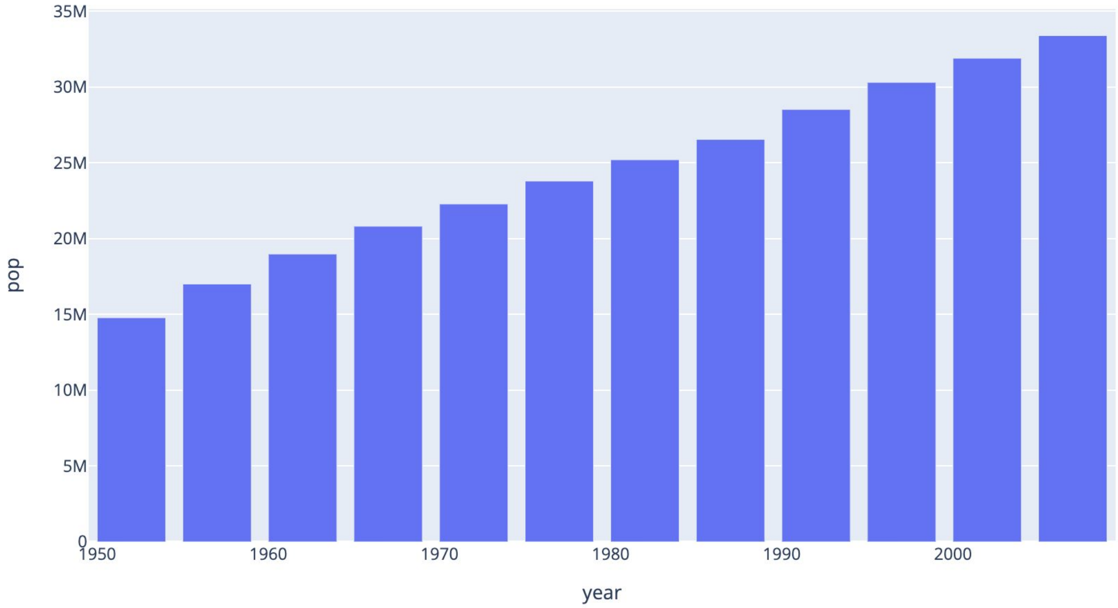
Color

Shape

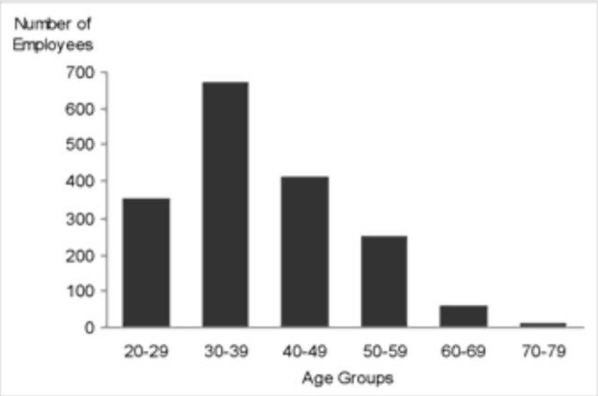
Kinds of categorical data - Nominal

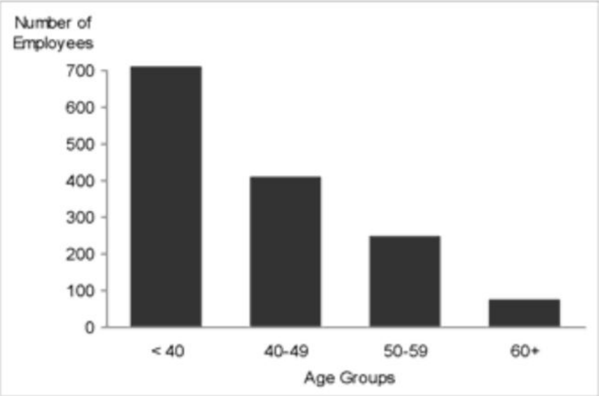


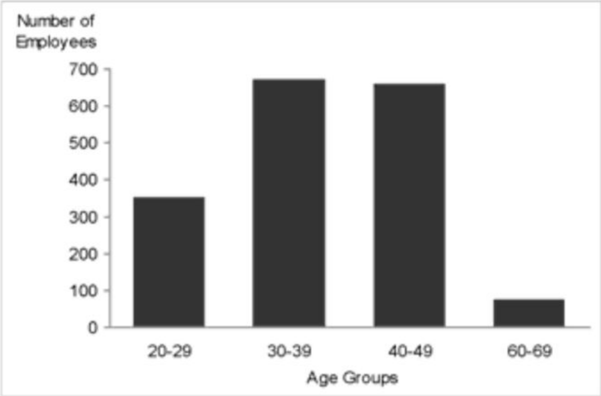
Kinds of categorical data - Ordinal



Kinds of categorical data - Interval



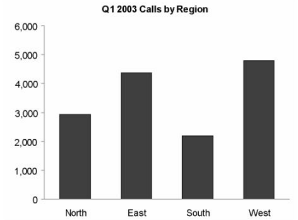
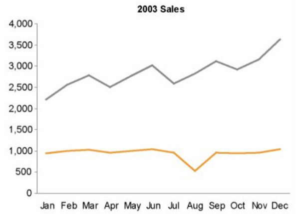
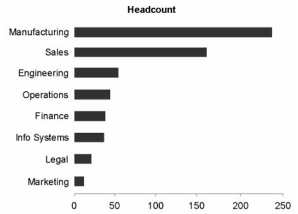
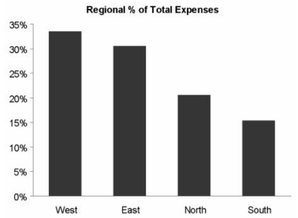




Can you spot the problem above?

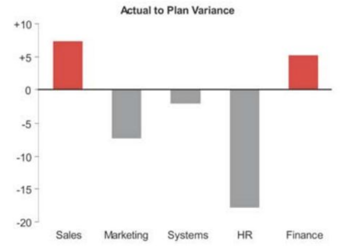
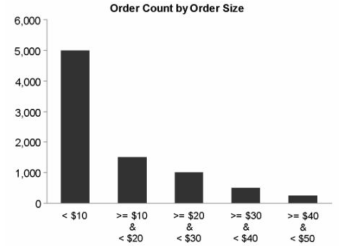
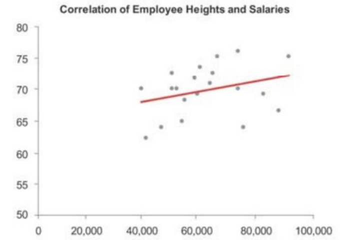
Kinds of graphs

[Eenie, Meenie, Minie, Moe: Selecting the Right Graph for Your Message]

Type/Description	Encoding Methods	Example
Nominal Comparison A simple comparison of the categorical subdivisions of one or more measures in no particular order	• Bars only (horizontal or vertical)	
Time Series Multiple instances of one or more measures taken at equidistant points in time	• Lines to emphasize overall pattern • Bars to emphasize individual values • Points connected by lines to slightly emphasize individual values while still highlighting the overall pattern • Always place time on the horizontal axis	
Ranking Categorical subdivisions of a measure ordered by size (either descending or ascending)	• Bars only (horizontal or vertical) • To highlight high values, sort in descending order • To highlight low values, sort in ascending order	
Part-to-Whole Measures of individual categorical subdivisions as ratios to the whole	• Bars only (horizontal or vertical) • Use stacked bars only when you must display measures of the whole as well as the parts	

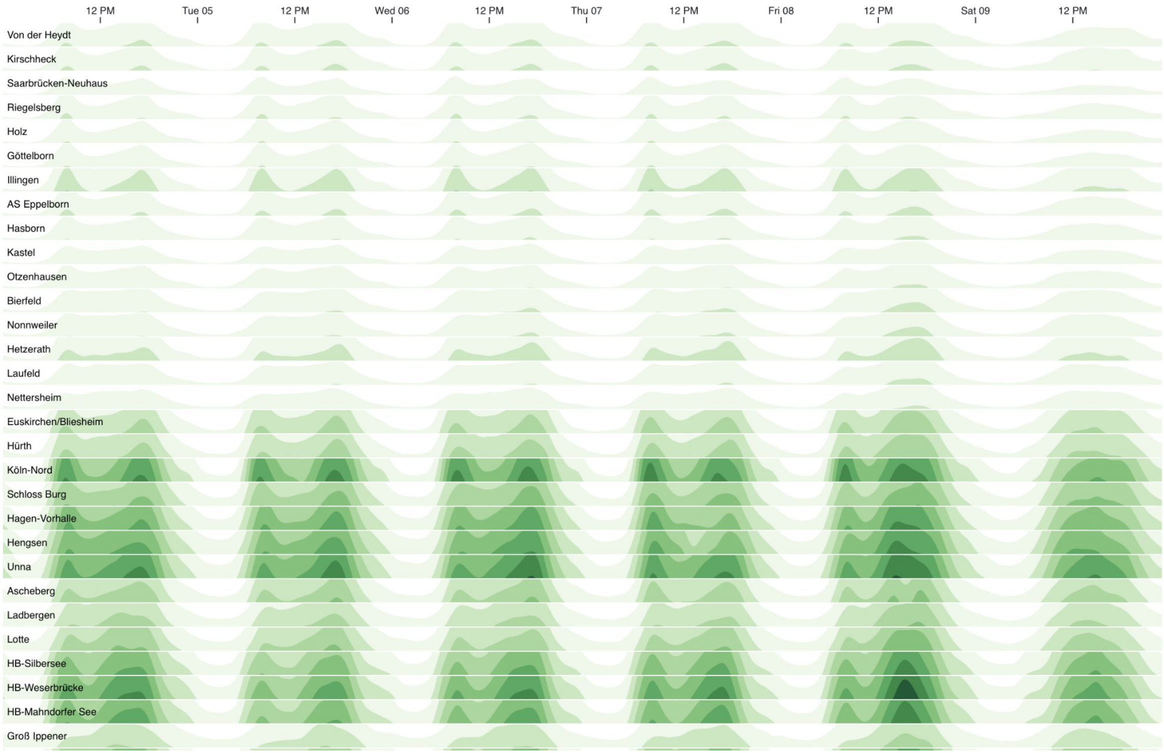
Kinds of graphs

[Eenie, Meenie, Minie, Moe: Selecting the Right Graph for Your Message]

Deviation Categorical subdivisions of a measure compared to a reference measure, expressed as the differences between them	• Lines to emphasize the overall pattern only when displaying deviation and time-series relationships together • Points connected by lines to slightly emphasize individual data points while also highlighting the overall pattern when displaying deviation and time-series relationships together • Bars to emphasize individual values, but limit to vertical bars when a time-series relationship is included • Always include a reference line to compare the measures of deviation against	
Frequency Distribution Counts of something per categorical subdivisions (intervals) of a quantitative range	• Vertical bars to emphasize individual values (called a *histogram*) • Lines to emphasize the overall pattern (called a *frequency polygon*)	
Correlation Comparisons of two paired sets of measures to determine if as one set goes up the other set goes either up or down in a corresponding manner, and if so, how strongly	• Points and a trend line in the form of a scatter plot • Bars may be used, arranged as a *paired bar graph* or a *correlation bar graph*, if scatter plots are unfamiliar • (Note: For descriptions of these graphs, see my book *Show Me the Numbers*.)	

Putting it all together

This is a good visualization - but can you spot what ‘rule’ it is breaking?



Take home message

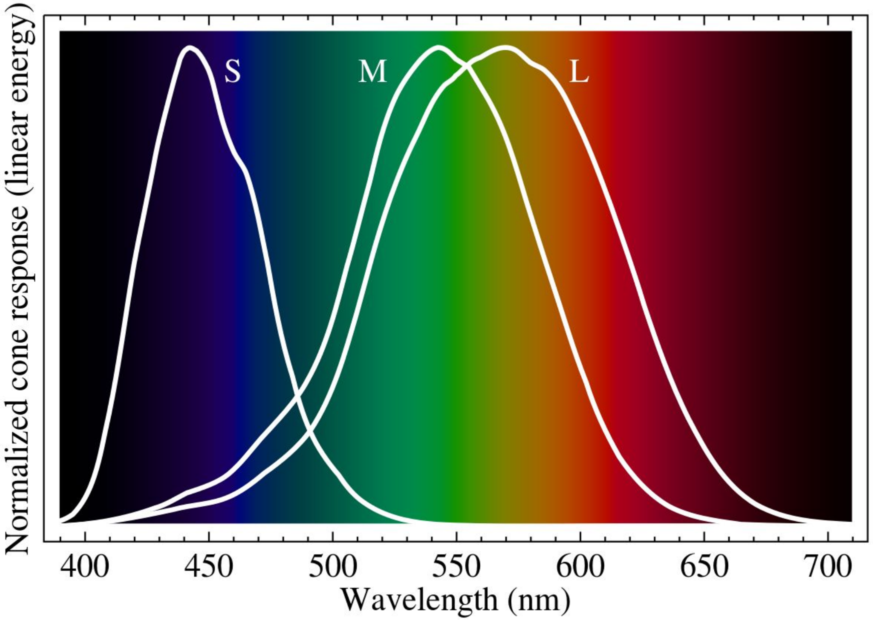
Visualization might be hard, but it’s not complicated

You should know your mapping from data space to visualization space and it should be one-to-one

Use your associations

Colors

What is color?



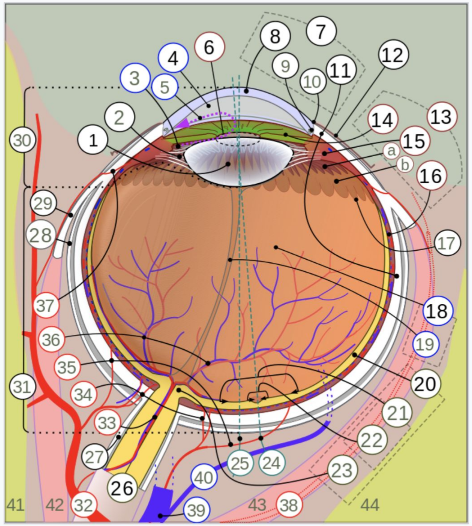


Diagram of a **human eye** (horizontal section of the right eye)

1. Lens, 2. Zonule of Zinn or Ciliary zonule, 3. Posterior chamber and 4. Anterior chamber with 5. Aqueous humour flow; 6. Pupil, 7. Corneosclera or Fibrous tunic with 8. Cornea, 9. Trabecular meshwork and Schlemm's canal. 10. Corneal limbus and 11. Sclera; 12. Conjunctiva, 13. Uvea with 14. Iris, 15. Ciliary body (with a: *pars plicata* and b: *pars plana*) and 16. Choroid); 17. Ora serrata, 18. Vitreous humor with 19. Hyaloid canal/(old artery), 20. Retina with 21. Macula or macula lutea, 22. Fovea and 23. Optic disc → blind spot; 24. Optical axis of the eye. 25. Axis of eye, 26. Optic nerve with 27. Dural sheath, 28. Tenon's capsule or bulbar sheath, 29. Tendon.

30. Anterior segment, 31. Posterior segment.

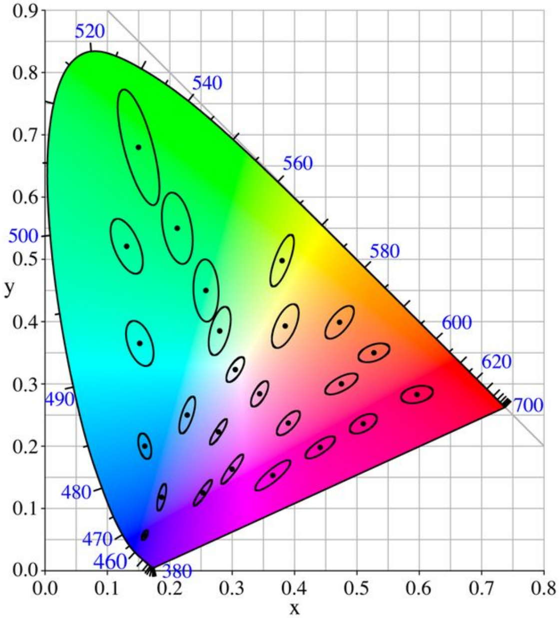
32. Ophthalmic artery, 33. Artery and central retinal vein → 36. Blood vessels of the retina; Ciliary arteries (34. Short posterior ones, 35. Long posterior ones and 37. Anterior ones), 38. Lacrimal artery, 39. Ophthalmic vein, 40. Vorticose vein.

41. Ethmoid bone, 42. Medial rectus muscle, 43. Lateral rectus muscle, 44. Sphenoid bone.

How far apart are two colors

How far apart are two colors?

Well…that depends on what space you are in



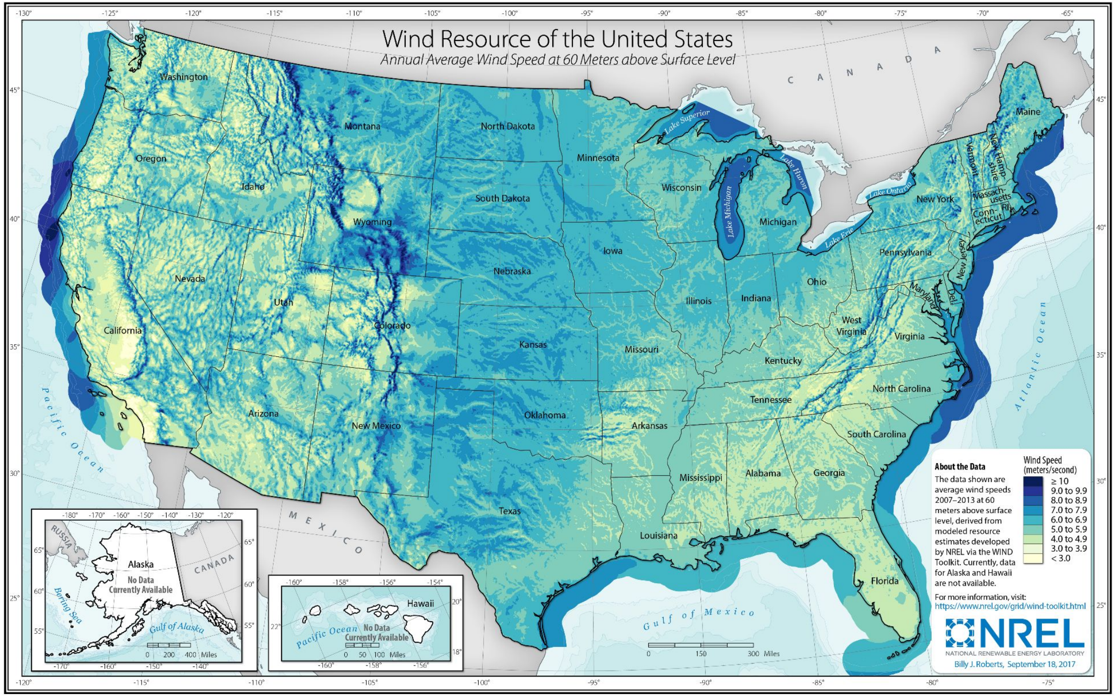
Macadam elipse

Color's two useful dimensions

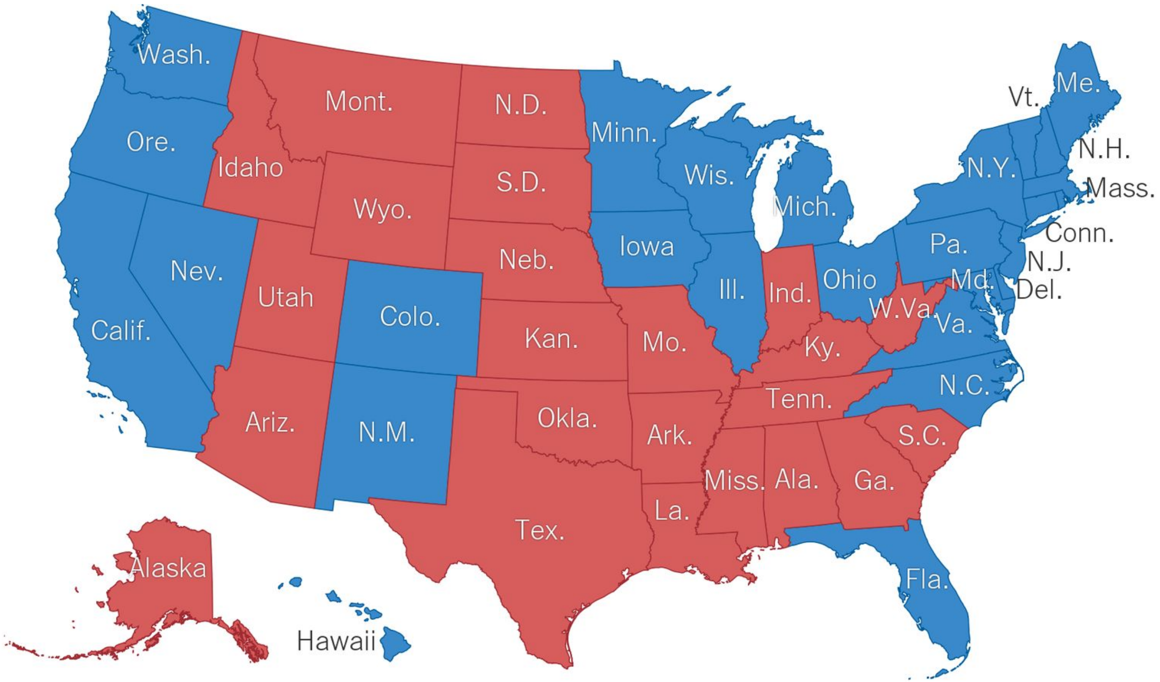
Saturation

Hue

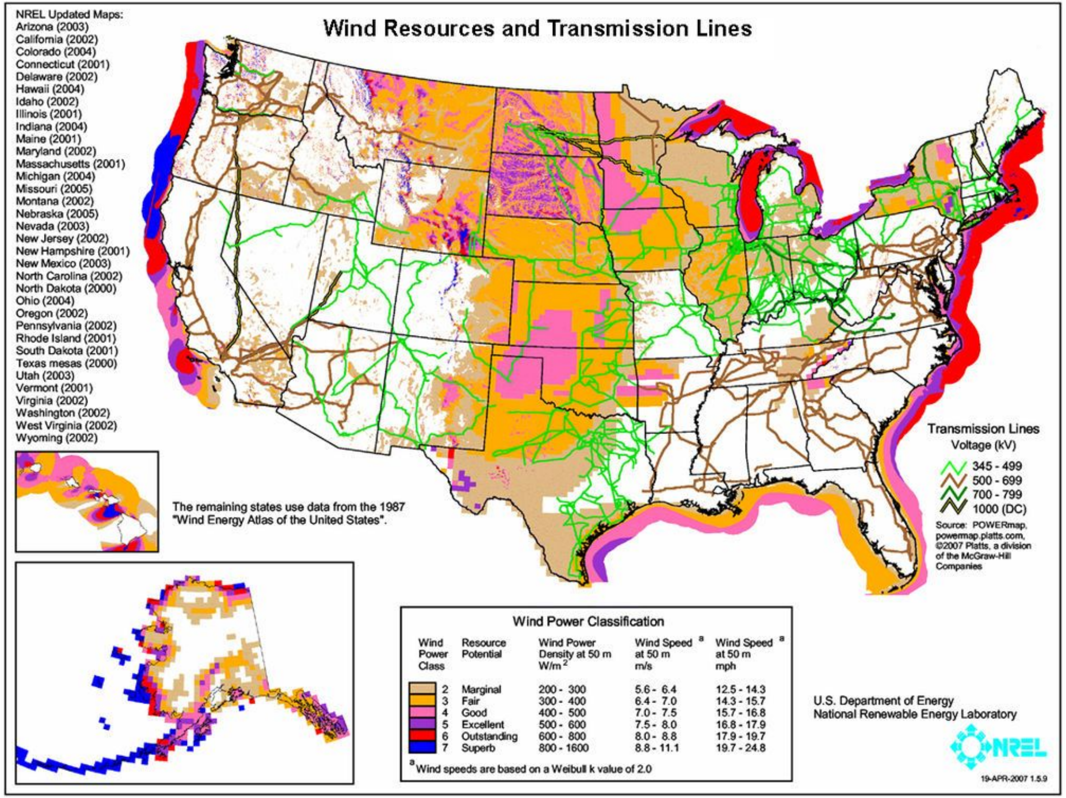
Saturation is for quantitative



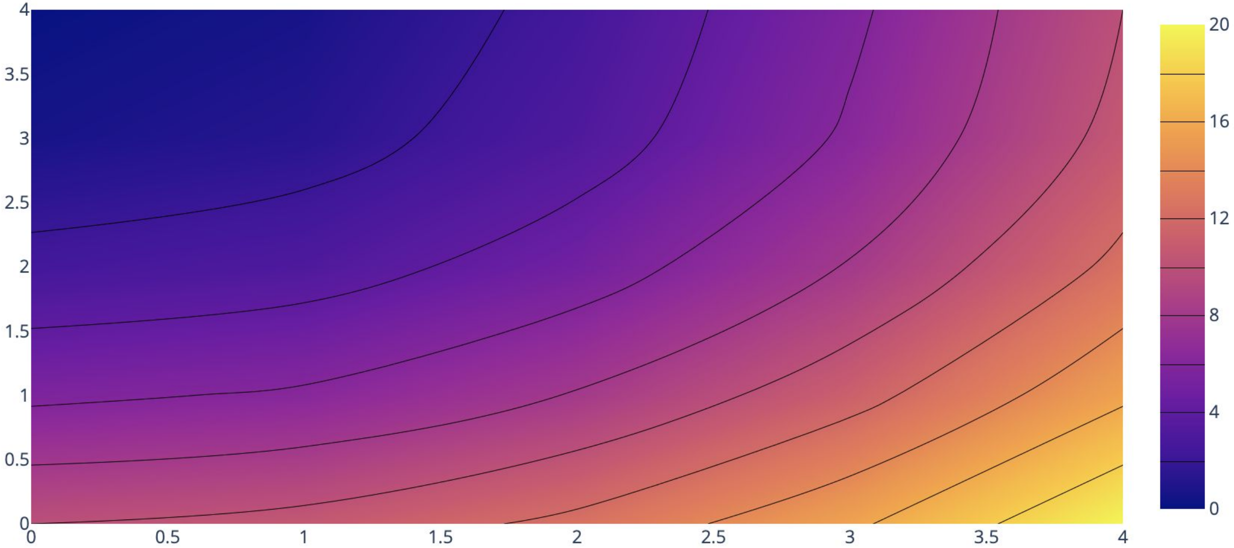
Hue is for categorical



What happens if you do not this



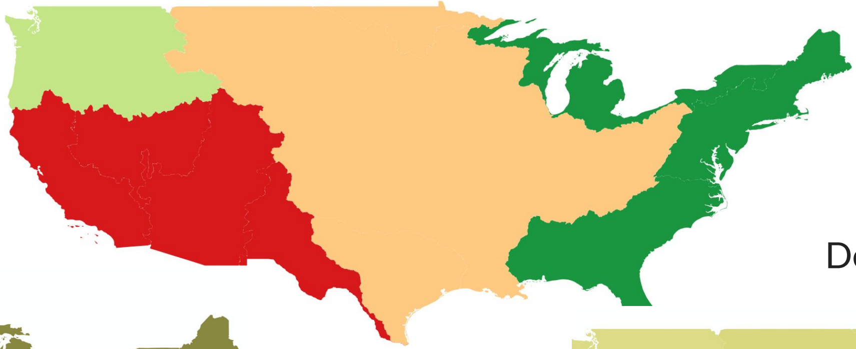
Can use two different hues for quantitative if needed



Colorblindness is real.

1. **Red Green Colorblindness**
    a. **Deuteranomaly**
        i. Red green colorblindness where red becomes more green
    b. **Protanomaly**
        i. Red green color blindness where red becomes more green and less bright
2. **Blue Yellow Colorblindess**
    a. **Tritanomaly**
        i. Hard to tell difference between blue/ green and yellow/red
    b. **Tritanopia**
        i. Hard to tell difference between blue/green, purple/red, and yellow /pink

Why are Colorblind Palettes Important?

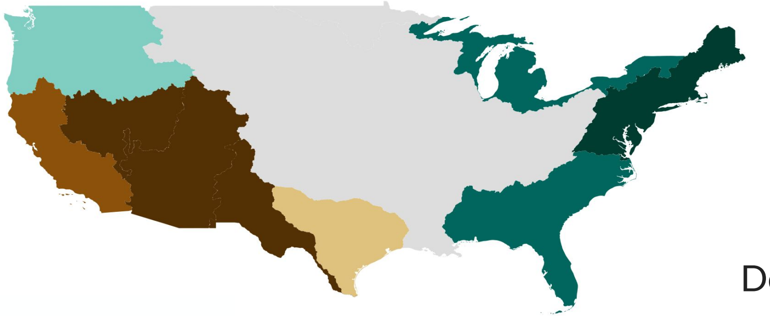


Original

Protanomaly

Deuteranomaly

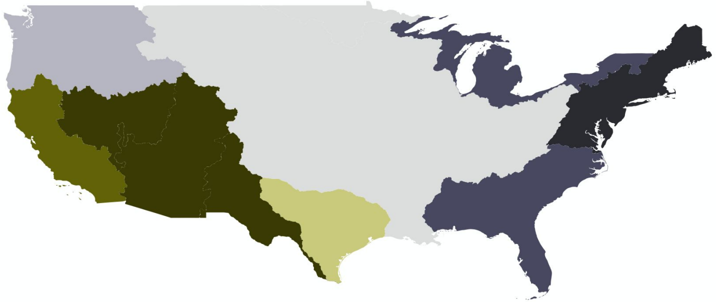
Why are Colorblind Palettes Important?



Protanomaly

Deuteranomaly

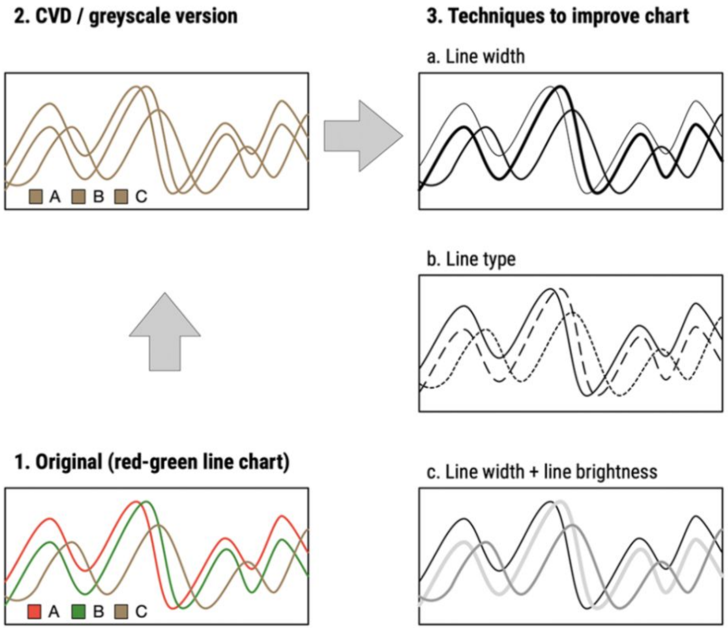
Original



Consider not using hue to avoid printing/display issues

From

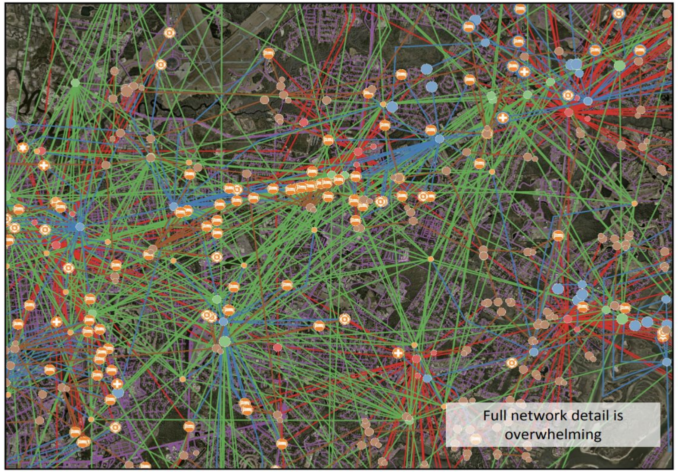
Stoelzle, Michael & Stein, Lina. (2021). Rainbow color map distorts and misleads research in hydrology – guidance for better visualizations and science communication. Hydrology and Earth System Sciences. 25. 4549-4565. 10.5194/hess-25-4549-2021.

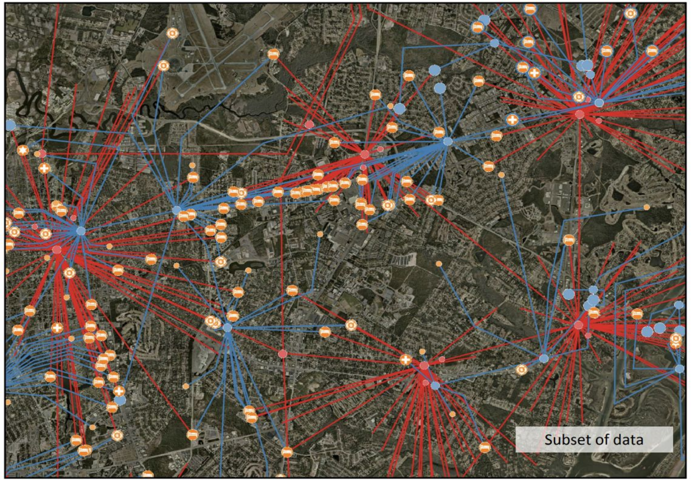


Some best practices from web design

Keep the information density as low as you can

“It is not how much information there is, but rather how effectively it is arranged." ~ Edward Tufte

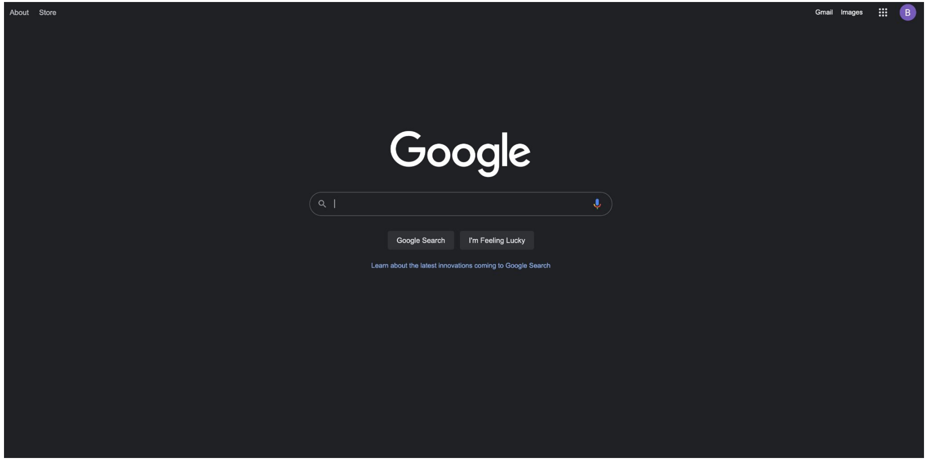




From: Multi-User Non-Linear Adaptive Magnification for Satellite Imagery and Graph Networks Sean Kim, Masters Thesis, RPI

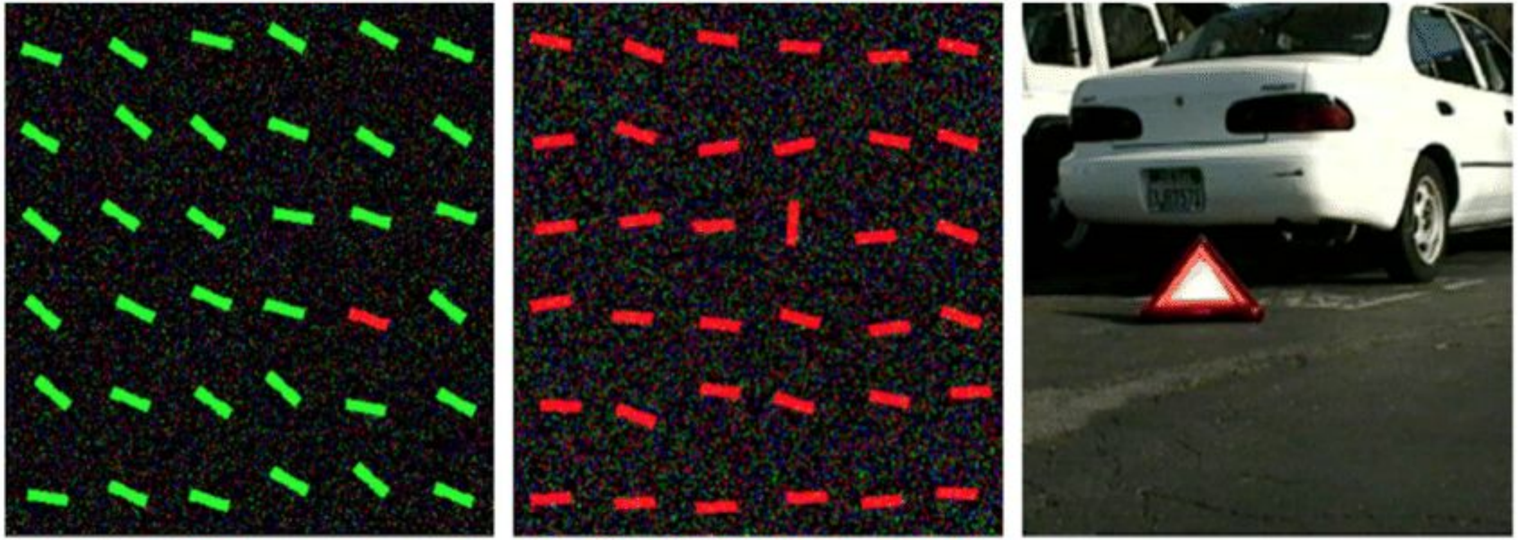
“Powerpoint is evil” ~ also Edward Tufte

This can be cultural

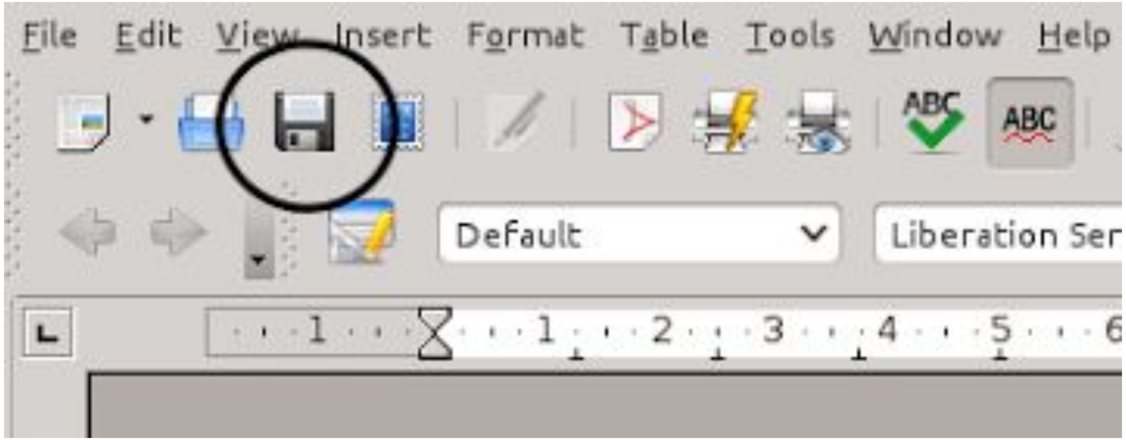


Keep it high contrast and use that to direct the eye

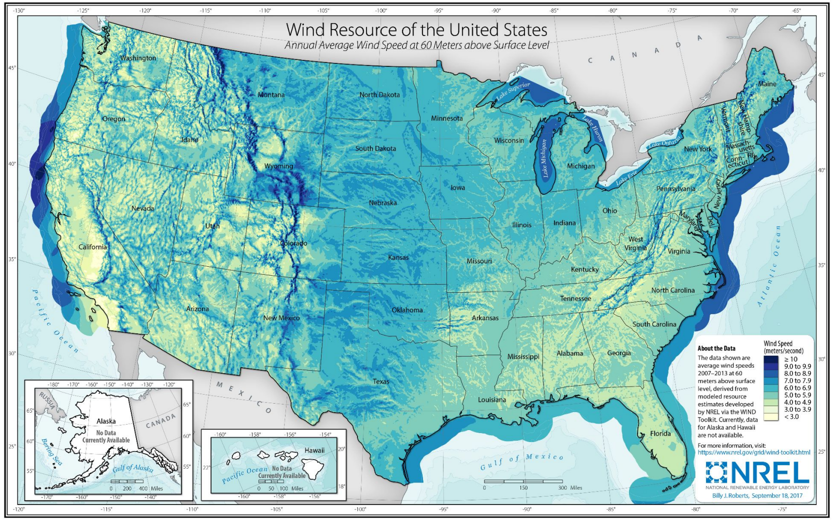
From Bors, Adrian & Papushoy, Alex. (2017). Image Retrieval Based on Query by Saliency Content. 10.1007/978-3-319-57687-9_8.

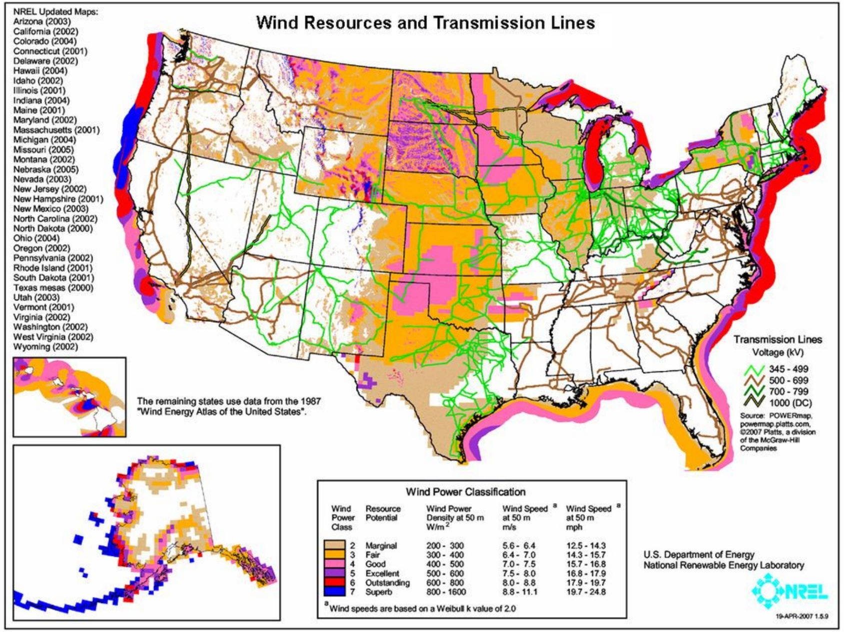


When in doubt, stick with the standard

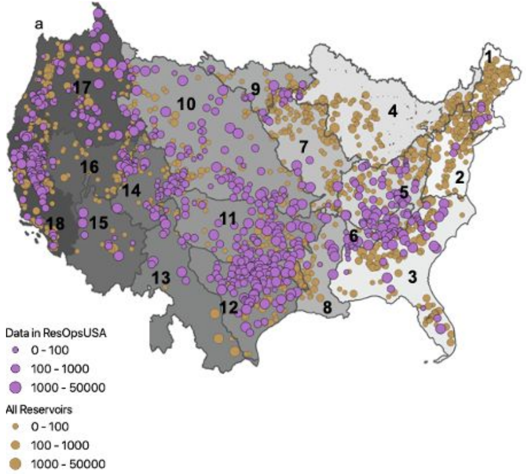


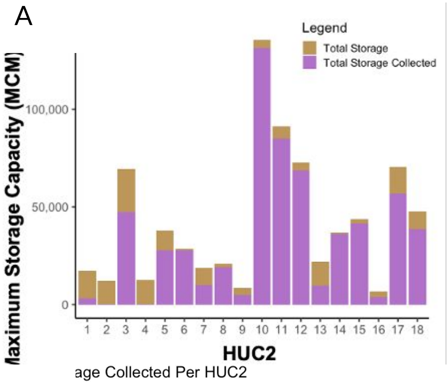
Recognition rather than recall

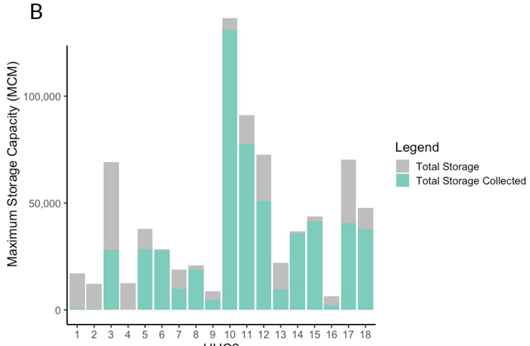




Consistent Colors Create Cohesiveness (C^4)

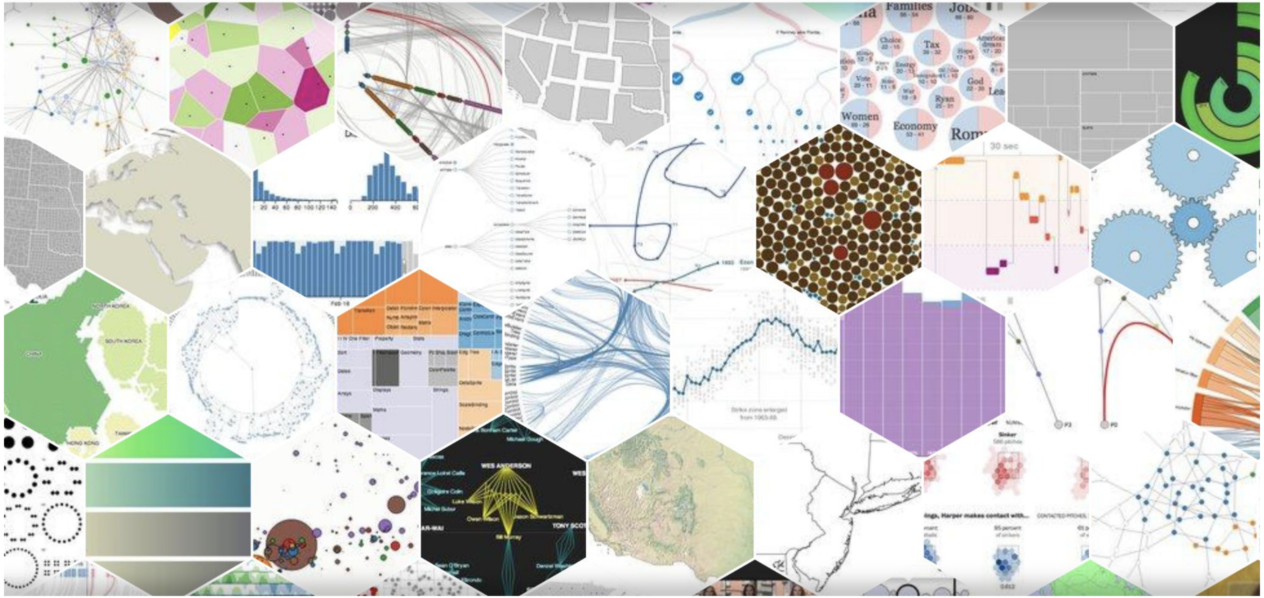






Take inspiration from others

Data-Driven Documents

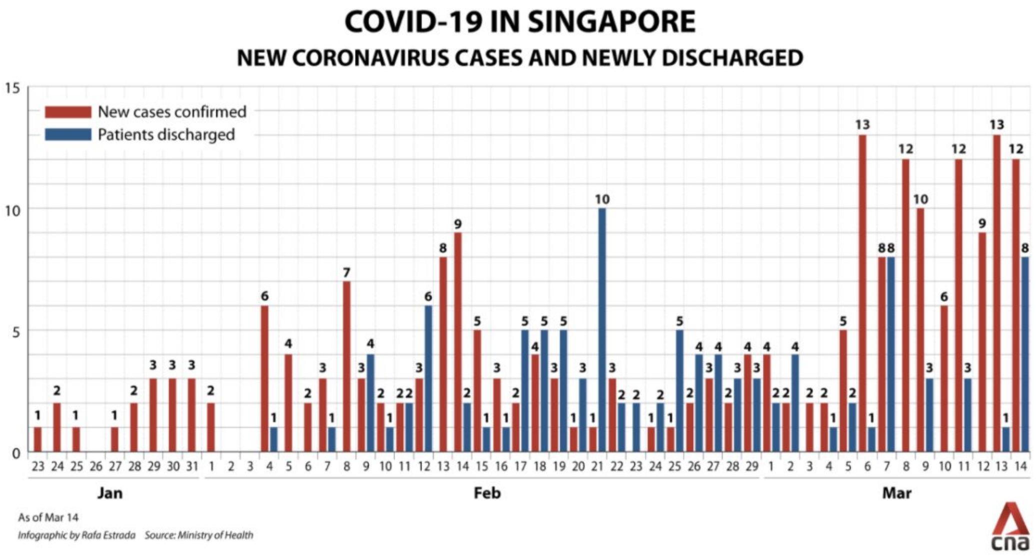


In summary

- Visualization is communication!!!
- Know your visualization ingredients
- What kind of data do you have?
- Know your mapping from one to another
- Saturation vs. Hue
- Low information density
- High contrast
- Use standards
- Build on intuition

Discussion

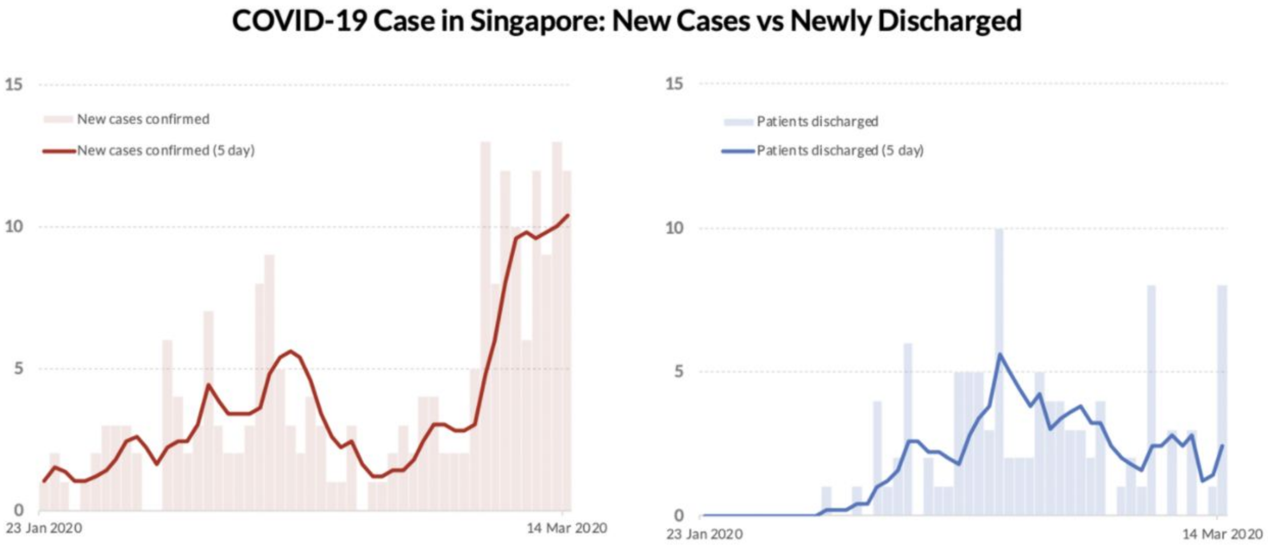
Good or bad? Why?



Source: Channel News Asia

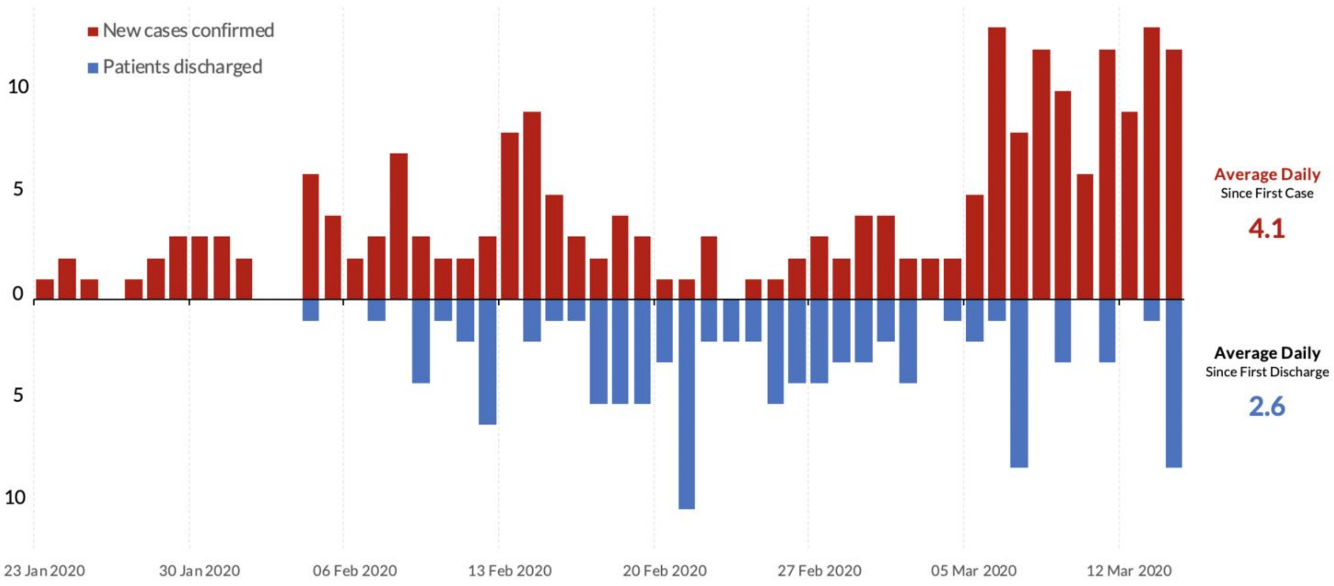
Ways to fix this

From analythical

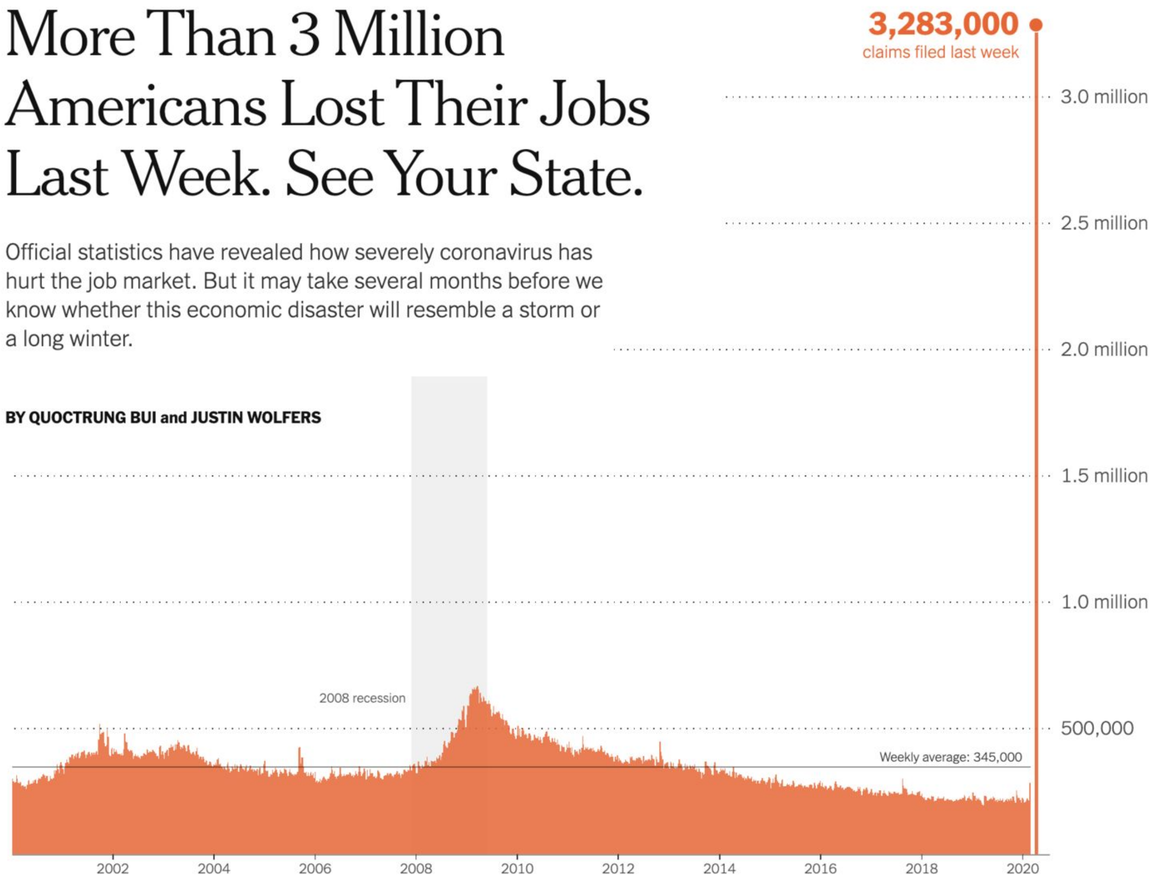


Covid-19 Singapore New Cases vs Discharged: Split Bar Chart with Moving Average

COVID-19 Case in Singapore: New Cases vs Newly Discharged



Good or bad? Why?



More Than 3 Million Americans Lost Their Jobs Last Week. See Your State.

Official statistics have revealed how severely coronavirus has hurt the job market. But it may take several months before we know whether this economic disaster will resemble a storm or a long winter.

BY QUOCTRUNG BUI and JUSTIN WOLFERS

Note: Official figures are seasonally adjusted. Source: Department of Labor

Lastly, some tools

Plot.ly for nice looking plots matplotlib style

ColorBrewer: https://colorbrewer2.org/#type=qualitative&scheme=Set1&n=8

Ggplot2 in R Studio

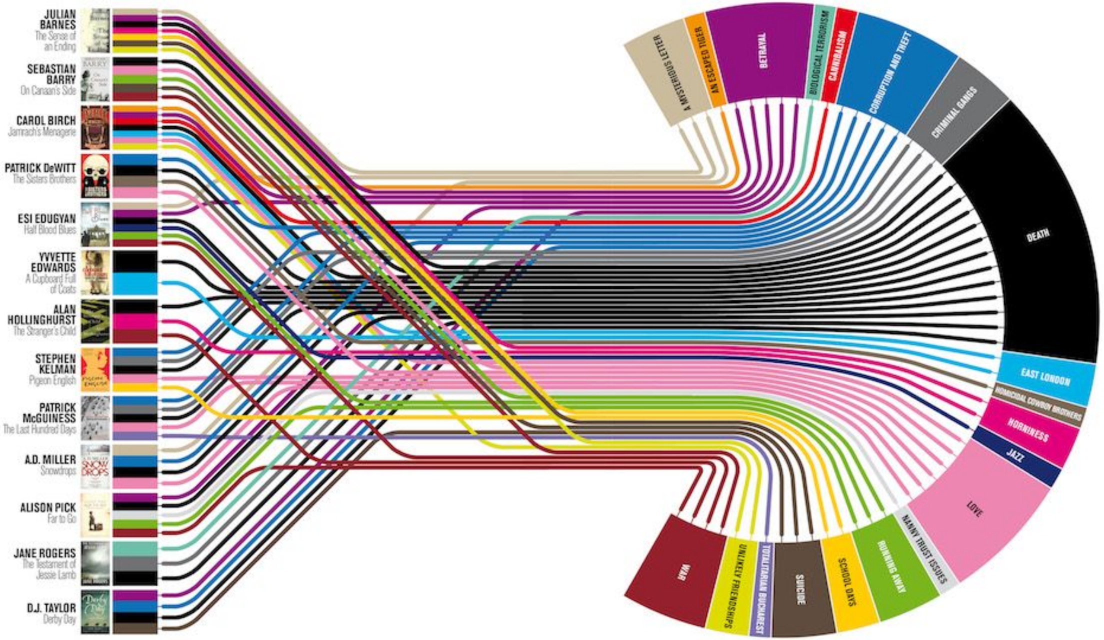
D3.js for getting fancy

Post Processing: Powerpoint

Other Resources/ Good Examples

https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1003833

Good or bad? Why?



Plot lines

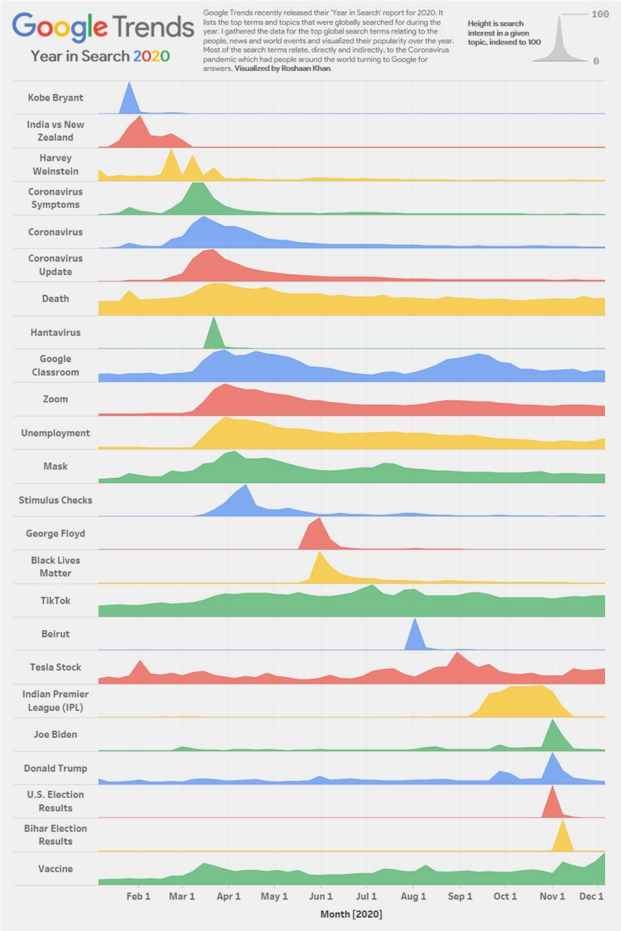
What makes a prize-winning novel? As Julian Barnes wins the Booker Prize, Delayed Gratification's Johanna Kamradt charts the themes of this year's longlisters.

Illustration: Christian Tate

Million dollar blocks

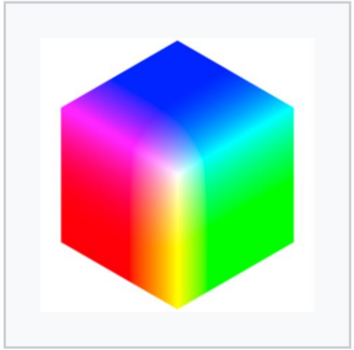
https://chicagosmilliondollarblocks.com/

Good or bad?
Why?



Some common color spaces

CIE 1931



BT.2020

DCI-P3

BT.709